

Principles of Economics

Problem Set 11

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

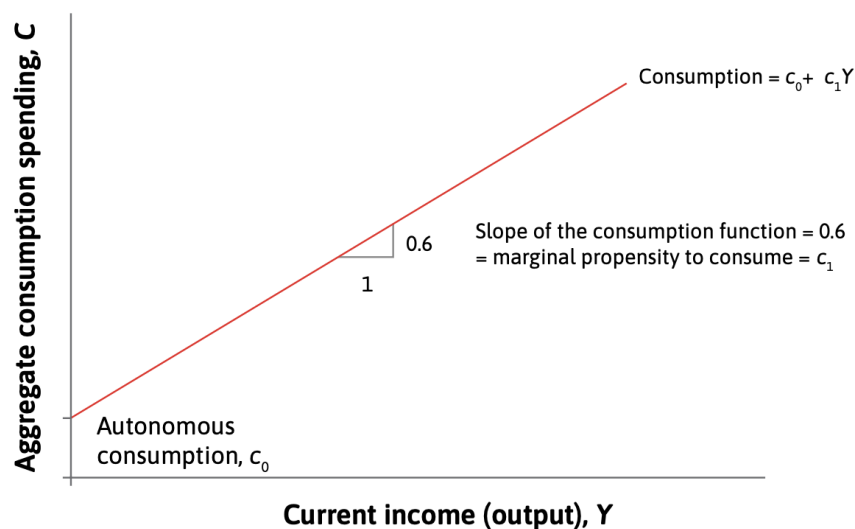
1. In France and Germany, it is difficult for a household to increase its borrowing based on an increase in the market value of the house. In addition, large down-payments (as a percentage of the house price) are required for house purchases. On the basis of this information, how would you expect a rise in house prices in France or Germany to affect spending by households?
2. In the US or UK, loans are more easily available based on a rise in home equity and only a small down-payment is required. How would you expect your answer to question 1 to change when considering the US or UK? What do you conclude about the role of the financial accelerator in France and Germany compared with the UK and the US?

2. MULTIPLE-CHOICE QUESTION

3. Figure 14.2 depicts a consumption function of an economy, where C is the aggregate consumption spending and Y is the current income of the economy.

Based on this information, which of the following statements is correct?

Figure 14.2



- a) The marginal propensity to consume (MPC) is the proportion of current income spent on consumption, C/Y .

- b) The MPC is given by the line's intercept on the vertical axis.
- c) The MPC is normally less than 1 as some households are able to smooth their consumption.
- d) If the current income of a country is $Y = \$100$ trillion and the $MPC = 0.6$, then the aggregate consumption spending is $C = \$60$ trillion.
4. The diagram depicts a consumption function of an economy, where C is the aggregate consumption spending, Y is the current income of the economy, and c_0 is the fixed (or autonomous) consumption such that $c_0 > 0$. Assume that households that are not credit-constrained would completely smooth their consumption. Which of the following statements is correct?
- If all households were not credit-constrained, and all income changes were perceived to be temporary, then the aggregate consumption line would be horizontal.
 - During a credit crunch when the banks become less willing to lend, the aggregate consumption line would become flatter.
 - If a higher proportion of households have "weakness of will", then the aggregate consumption line would be flatter.
 - If the current income falls to zero, there will be zero consumption.
5. Assuming that there is no government spending or trade, an economy's aggregate demand is given by its domestic consumption C and investment I , $AD = C + I = c_0 + c_1 Y + I$. In the economy's goods market equilibrium this equals its output: $AD = Y$. Solving for Y this yields:
- $$Y = [1/(1 - c_1)] (c_0 + I)$$
- Given this equation, which of the following statements is correct?
- The multiplier is given by $1 - c_1$.
 - The boost in the economy's output is the same, regardless of whether the aggregate demand shock comes from an increase in investment I or in autonomous consumption c_0 .
 - The larger the marginal propensity to consume (c_1), the smaller the multiplier.
 - If $c_1 = 1/3$, then a £1 million increase in investment would result in a £2 million increase in output, *ceteris paribus*.
6. In the US and the UK, loans are widely available based on a rise in home equity. Additionally, unlike in France and Germany where large down-payments (as a percentage of the house price) are required, in the US and the UK only small down-payments are required for house purchases. On the basis of this information, which of the following statements is correct for the US and the UK when there is a rise in housing prices.
- There is a positive financial accelerator effect for the existing homeowners who are credit-constrained.
 - There would be no effect on the consumption of existing homeowners who are not credit-constrained.
 - Aspiring homeowners are likely to increase saving and reduce their consumption more than they would in France and Germany.
 - A rise in housing prices is likely to dampen consumption in the US and the UK.

7. Which of the following statements regarding the multiplier is correct?
- a. If two countries were identical except for the share of credit-constrained households, then the country with the higher share would have a smaller multiplier.
 - b. The multiplier is constant over the business cycle.
 - c. An increase in the level of exports leads to a higher multiplier.
 - d. Taxation and imports are “leakages” from the circular flow of income, which reduce the size of the multiplier.

3. NUMERICAL PROBLEMS

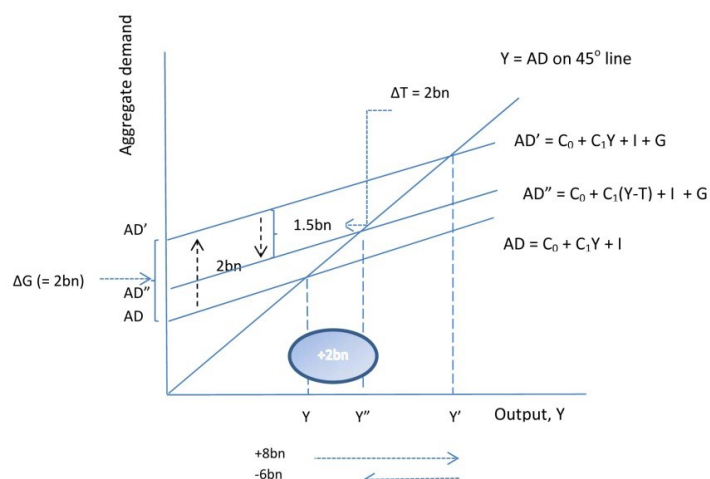
8. Assume the economy is in a recession. The government has a high level of debt and wants to set a balanced budget, that is, $G = T$. How can the government achieve a fiscal stimulus effect on GDP whilst keeping the budget balanced?

To answer the question, take the following steps:

- a. Show how this is possible in a multiplier diagram, ensuring that you label the relevant intercepts and angles. Make the diagram sufficiently accurate so that the exact size of the multiplier is visible.
- b. Explain in words how the government can achieve such a fiscal stimulus effect whilst keeping the budget balanced.
- c. Derive the balanced budget multiplier using algebra. (Hint: You will need to write down expressions for the change in GDP associated with a change in both G and T and set these equal.)
- d. Comment briefly on any disadvantages you see with the use of this balanced budget fiscal stimulus.

You can make the following assumptions:

- a. Assume a lump sum tax. This means that the tax does not depend on the level of income, $T = T$, rather than our usual assumption that $T = tY$.
- b. Also assume that the country does not have any imports or exports.



9. Consider the multiplier model

- Compare two economies, which differ only in their share of credit constrained households but are identical otherwise. In which economy is the multiplier larger? Illustrate your answer using a diagram.
- On the basis of your comparison of the two economies, would you expect the multiplier in an economy to vary over its business cycle?
- Some economists estimated the size of the multiplier in the Great Depression to be equal to 1.8. Explain how the following characteristics of the US economy at the time could have affected its value:
 - the size of government (see Figure 14.1)
 - the fact that there were no unemployment benefits
 - the fact that the share of imports was small

$$k = \left(\frac{1}{1 - c_1(1 - t) + m} \right)$$

4. CONCEPTUAL QUESTIONS

- Consider the three methods discussed in this unit that have been used to estimate the size of the multiplier: the Mafia-related dismissals in Italy, the stimulus highway spending in the US, and wartime defence spending in the US. Why do you think estimates of the size of the multiplier vary? Use the material in this unit to support your explanation.
- In an article from August 2014, 'The Fall of France' (<http://tinyco.re/7111032>), Paul Krugman criticizes the austerity policy implemented in France. Use what you have learned about the fiscal multiplier to explain why, in Krugman's opinion, fiscal austerity in France (and more generally in Europe) would fail (explain carefully what you think Krugman means by 'fail').